HOMEWORK 7
An Information Theoretic Algorithm

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An Information Theoretic Algorithm for Finding Periodicities in Stellar Light Curves

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***Abstract*—We propose a new information theoretic metric for finding periodicities in stellar light curves. Light curves are astronomical time series of brightness over time, and are characterized as being noisy and unevenly sampled. The proposed metric combines correntropy (generalized correlation) with a periodic kernel to measure similarity among samples separated by a given period. The new metric provides a periodogram, called Correntropy Kernelized Periodogram (CKP), whose peaks are associated with the fundamental frequencies present in the data. The CKP does not require any resampling, slotting or folding scheme as it is computed directly from the available samples. CKP is the main part of a fully-automated pipeline for periodic light curve discrimination to be used in astronomical survey databases. We show that the CKP method outperformed the slotted correntropy, and conventional methods used in astronomy for periodicity discrimination and period estimation tasks, using a set of light curves drawn from the MACHO survey. The proposed metric achieved 97.2% of true positives with 0% of false positives at the confidence level of 99% for the periodicity discrimination task; and 88% of hits with 11.6% of multiples and 0.4% of misses in the period estimation task.**

***Index Terms*—Correntropy, information theory, period detection, period estimation, time series analysis, variable stars.**

analysis is an important tool in astrophysics used for estimation of stellar masses and distances to astronomical objects. By analyzing the light curves derived from the sky surveys, astronomers can perform tasks such as transient event detection, variable star detection and classification.

There are a certain types of variable stars [1] whose brightness varies following regular cycles. Examples of this kind of stars are the pulsating variables and eclipsing binary stars. Pulsating stars, such as Cepheids and RR Lyrae, expand and contract periodically effectively changing their size, temperature and brightness. Eclipsing binaries, are systems of two stars with a common center of mass whose orbital plane is aligned to Earth. Periodic drops in brightness are observed due to the mutual eclipses between the components of the system. Although most stars have at least some variation in luminosity, current ground based survey estimations indicate that 3% of the stars varying more than the sensitivity of the instruments and $\sim 1\%$ are periodic [2].

Detecting periodicity and estimating the period of stars is of high importance in astronomy. The period is a key feature

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